TURBULENCE THROUGH CANONICAL FLOWS

Transition, spectra, examples, data, and computation

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Fluid Dynamics and IBM Theory • Module 5

Adapted from a lecture by Steven L. Brunton.

Guiding idea

Canonical flows isolate mechanisms that recur in more complicated systems.

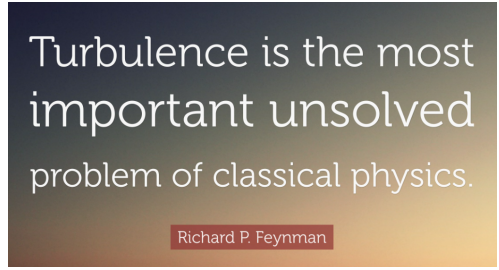
Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

LECTURE ROADMAP

- Why turbulence is hard (and why it matters)
- Canonical flows: a “vocabulary” for turbulence
- Examples: pipe, wake, boundary layer, cavity, jets, mixing layers
- “More complex” flows (geometry, separation, multi-scale physics)
- Open-source resources: databases and CFD codes

WHY TURBULENCE IS HARD

- Turbulence is a *multiscale*, nonlinear dynamics problem whose observed state can depend on disturbance history.
- Small perturbations can grow, interact, and reorganize the flow.
- Many flows are *linearly stable* yet become turbulent through finite-amplitude disturbances.
- Predicting transition and statistics (drag, mixing, heat transfer, intermittency) is still challenging.



Quotation graphic (stock “quote image”; originating website to be confirmed). The wording is a popular paraphrase attributed to R. P. Feynman; his own text says turbulence is an old, unsolved problem common to many fields (*Feynman Lectures on Physics*, Vol. I, Ch. 3).

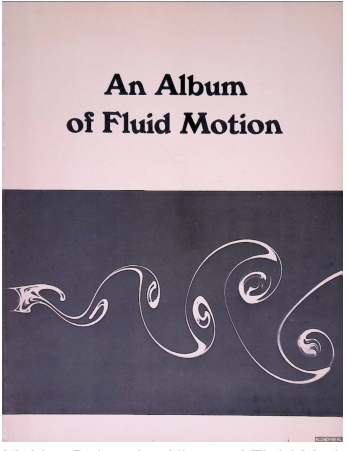
The *Feynman Lectures* (Vol. II, Ch. 41) give a classic discussion of why real viscous flows are difficult to predict.

CANONICAL FLOWS (VAN DYKE “ALBUM”)

Idea

A small set of “canonical” configurations show up *everywhere*. Learn these and you gain a language for:

- transition mechanisms (instability $\rightarrow$ breakdown),
- coherent structures (vortices, streaks, shear layers),
- scaling laws and nondimensional groups.



M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982), cover.

A MINIMAL TOOL: REYNOLDS NUMBER

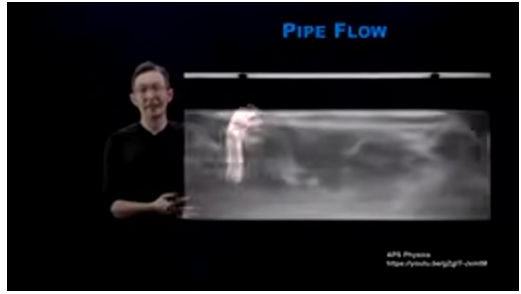
- For chosen velocity and length scales, Reynolds number compares inertia to viscosity:

$$\mathrm{Re} = \frac{UL}{\nu}.$$

- "High" Re often permits thinner shear layers and wider scale separation, but does not guarantee turbulence.
- Treat Re as a control parameter, not a universal one-to-one label for complexity: geometry, forcing, roughness, and disturbance history matter.

PIPE FLOW

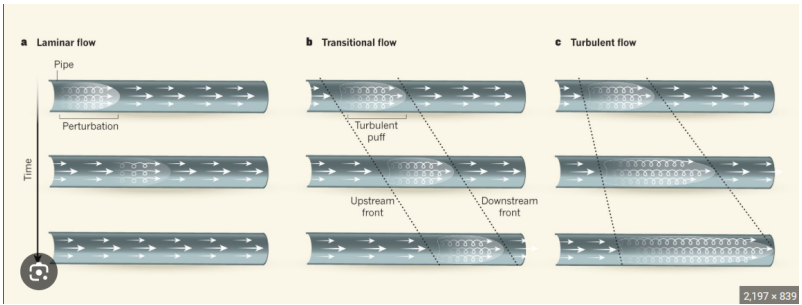
- Classic internal flow: long straight pipe, pressure-driven.
- The Hagen–Poiseuille solution exists for every Re and is linearly stable (proved for axisymmetric disturbances; computed stable to $\mathrm{Re} = 10^7$, Meseguer & Trefethen 2003), yet finite-amplitude disturbances can trigger turbulence. (Their Re uses centerline speed and radius, which for the parabolic profile equals $U_b D/\nu$.)
- Transition often involves localized “puffs” and “slugs” (intermittency).
- Active control is possible: carefully designed perturbations can suppress turbulence (drag reduction).



Still from S. L. Brunton’s canonical-flows lecture (linked on the title slide); the embedded pipe-flow dye visualization is credited on screen to an APS Physics video on YouTube (full URL to be confirmed).

PIPE FLOW: SLUGS AND PUFFS

- Transitional pipe flow can coexist with laminar regions and is intrinsically intermittent.
- Localized turbulent structures have distinct dynamics:
  - **puffs** remain localized while advecting and may decay or split stochastically;
  - at higher Re, **slugs** expand by turbulent invasion at their fronts.
- The crossover is a range of behavior rather than a single universal threshold.



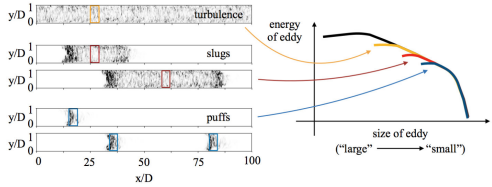
Source: Graham, M. Turbulence spreads like wildfire. Nature 526, 508–509 (2015).
M. D. Graham's News & Views commentary (doi:10.1038/526508a) on D. Barkley et al., "The rise of fully turbulent flow," *Nature* **526**, 550–553 (2015).

PUFFS, SLUGS, AND SPECTRAL SIMILARITY

- The image sketches spectra sampled from the marked pipe-flow regions.
- For high-Re, locally isotropic **3D** turbulence, Kolmogorov similarity hypothesizes an inertial range

$$E(k) = C_K \varepsilon^{2/3} k^{-5/3}.$$

- This is asymptotic and statistical—not exact universality. Intermittency, anisotropy, walls, forcing, and finite Re produce deviations.
- A 2D inverse-energy cascade may also have a $-5/3$ range, but its transfer direction and mechanism differ; enstrophy transfers downscale.



Source image: Phys.org (2020).

The news item reports work of the Chakraborty group (OIST) on spectra of puffs and slugs; see also R. T. Cerbus, C.-C. Liu, G. Gioia & P. Chakraborty, *Phys. Rev. Lett.* **120**, 054502 (2018). Original figure credit as given in the news item (to be confirmed).

WAKE FLOWS

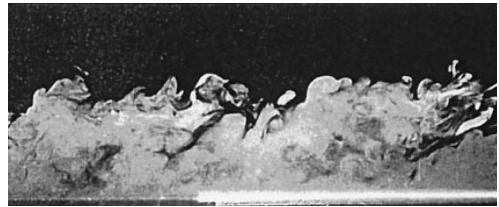
- Flow past a body leaves a *wake* with a velocity deficit and strong shear.
- Bluff bodies can shed a Kármán vortex street, producing unsteady loading and noise; lifting surfaces also generate trailing and tip vortices.
- Wakes are central to:
  - aircraft wing tips and propellers,
  - wind farms and arrays,
  - animal locomotion and swimming.



Still from S. L. Brunton's canonical-flows lecture (linked on the title slide); embedded wake images credited on screen to B. Herrmann, P. Oswald, R. Semaan & S. L. Brunton, *Commun. Phys.* (2020) ("Modeling synchronization in forced turbulent oscillator flows"; open access, CC BY 4.0; details to be confirmed).

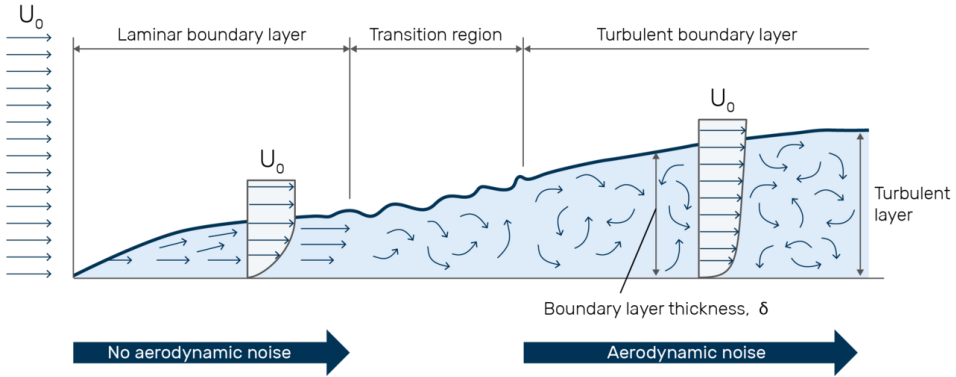
BOUNDARY LAYER

- Near a wall, no-slip forces a thin region of strong shear: the boundary layer.
- Key questions:
  - when does laminar flow transition to turbulence?
  - how do near-wall streaks and vortices sustain turbulence?
  - how do roughness and pressure gradients change drag?
- Boundary layers dominate drag and heat transfer on vehicles, buildings, and animals.



Smoke visualization of a turbulent boundary layer; Van Dyke, *An Album of Fluid Motion* (turbulent boundary layer plate; also reproduced in Dou 2018).
Photographer (likely R. E. Falco) and plate number to be confirmed. Note the sharp, convoluted edge between turbulent (smoke-filled) and irrotational outer fluid: boundary-layer intermittency.

BOUNDARY LAYER: SPATIAL DEVELOPMENT

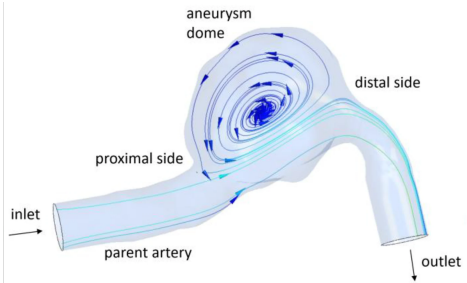


Boundary-layer schematic after GRAS Sound & Vibration, *The boundary layer—laminar and turbulent flows* (blog; source to confirm), https://www.grasacoustics.com/blog/aerodynamic-measurements-components-of-the-boundary-layer.

Reading the figure: the boundary-layer thickness δ grows as $\delta \propto x/\mathrm{Re}_x^{1/2}$ in the laminar region (Blasius) and faster, roughly $\delta \propto x/\mathrm{Re}_x^{1/5}$, once turbulent; on a smooth flat plate transition typically occurs for $\mathrm{Re}_x = U_0 x/\nu \sim 3\times10^5$–$3\times10^6$, depending on free-stream disturbances. The “no aerodynamic noise” label is a simplification: laminar layers radiate far less, not nothing.

CAVITY FLOW

- In an **open engineering cavity**, a shear layer spans the opening; pressure/acoustic feedback can create resonant oscillations.
- An intracranial aneurysm is instead a **cavity-like recirculating flow**: pulsatility, geometry, and vessel-wall conditions shape its hemodynamics.
- Related applications:
  - aneurysm inflow jets and recirculation,
  - gaps, steps, and surface defects,
  - flow-induced vibration.



Source: Boniforti MA, Magini R, Orosco Salinas T. Hemodynamic Investigation of the Flow Diverter Treatment of Intracranial Aneurysm. Fluids. 2023; 8(7):189.

JETS, MIXING LAYERS, AND SEPARATED FLOWS (OVERVIEW)

- **Round jet:** downstream of the potential core the mean profile becomes *self-similar*, the half-width grows roughly linearly, and Kelvin–Helmholtz roll-up of the annular shear layer seeds the initial vortex rings. The half-velocity radius grows as $\mathrm{d}r_{1/2}/\mathrm{d}x \approx 0.09$–$0.10$ (about 5–6°; Pope 2000, §5.1); the visible turbulent edge spreads at a half-angle of order 10–12°.
- **Plane mixing layer:** two streams of different speed produce the Brown–Roshko coherent spanwise rollers, which pair and merge; at high Re the thickness grows linearly, $\mathrm{d}\delta/\mathrm{d}x \propto \lambda = (U_1 - U_2)/(U_1 + U_2)$, and is essentially independent of Re (and only weakly dependent on the density ratio).

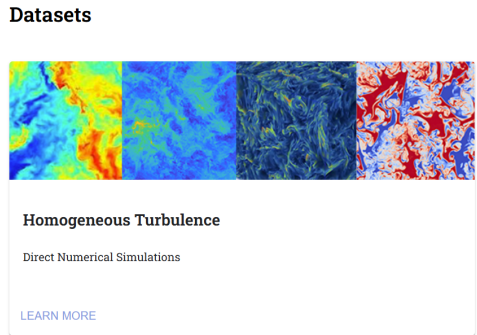
> **Self-similar** means one profile shape after rescaling: for the round jet $\bar{U}(r,x)/U_0(x) = f\big(r/r_{1/2}(x)\big)$, with centreline velocity $U_0(x) \propto 1/x$ and $r_{1/2} \propto x$, so the momentum flux $\propto U_0^2 r_{1/2}^2$ stays constant (Pope 2000, §5.1).

OVERVIEW, CONTINUED: SEPARATED AND COMBINED FLOWS

- **Separated flows:** a backward-facing step separates at a fixed geometric edge; at high Re the mean reattachment length is about 6–7 step heights (7 ± 1 in the review of Eaton & Johnston 1981, varying with expansion ratio and the incoming boundary layer; laminar steps reattach at an Re-dependent distance), and the shear layer, recirculation bubble, and reattachment zone each have their own statistics.
- **Combinations:** real configurations superpose these—a jet issuing into a cavity, a wake feeding a mixing layer, a step downstream of a boundary layer.
- **Multi-scale physics:** geometry sets the large scales; viscosity sets the small ones. The gap between them is what makes prediction expensive.

OPEN DATA: TURBULENCE DATABASES

- Public datasets let you test ideas without running a massive DNS yourself.
- Example: Johns Hopkins Turbulence Databases (JHTDB)
  - isotropic turbulence, channel flow, boundary layers, geophysical, . . .
  - query-able via web/API-style interfaces
  - great for teaching statistics, structure functions, spectra, and coherent-structure detection



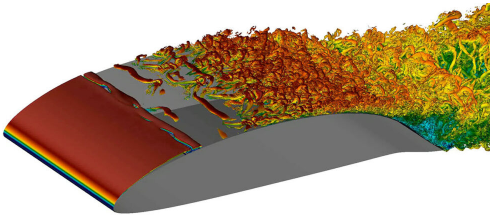
Johns Hopkins Turbulence Databases, turbulence.pha.jhu.edu.

OPEN-SOURCE CFD CODES

Two common ecosystems

- **OpenFOAM**: open-source finite-volume framework with many RANS/LES solvers.
- **Nek5000**: high-order spectral-element code; widely used for DNS/LES in complex geometries.

- Choosing a tool depends on: accuracy needs, geometry, turbulence model, and HPC resources.



Nek5000 project banner image, Argonne National Laboratory, nek5000.mcs.anl.gov.

TAKEAWAYS

- Turbulence is everywhere, but canonical flows give structure to the chaos.
- Each canonical flow highlights different mechanisms (instability, shear, walls, separation, resonance).
- Open videos, datasets, and codes make it easier than ever to *see* turbulence and to explore it quantitatively.

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- Official resources: JHTDB, OpenFOAM Foundation, and Nek5000.

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- S. L. Brunton, lecture on canonical turbulent flows (source of the slide sequence and of the two lecture stills), Cassyni seminar page.
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- R. T. Cerbus, C.-C. Liu, G. Gioia, and P. Chakraborty, "Kolmogorovian turbulence in transitional pipe flows," *Phys. Rev. Lett.* **120** (2018), 054502.
- M. A. Boniforti, R. Magini, and T. Orosco Salinas, "Hemodynamic investigation of the flow diverter treatment of intracranial aneurysm," *Fluids* **8** (2023), 189 (open access).
- Remaining image credits (Phys.org, GRAS, JHTDB, Nek5000) are on the slides where they appear.